Muhammad Hassaan Shahid

Electronics Engineer | P.E.C Reg. ELECT/112783

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SUMMARY

Electrical (Electronics) Engineer with strong foundation in industrial automation, control systems, PLC programming, and machine learning applications in biomedical engineering. Gained practical industry experience as Management Trainee Engineer at Volka Food International and hands-on exposure through internship at MEPCO. Published research paper in a reputed international journal (2025).

EDUCATION

- B.Sc. Electrical Engineering (Specialization: Electronics)
  NFC Institute of Engineering & Technology, Multan **2021 –2025**

PROFESSIONAL EXPERIENCE

Graduate Engineer Trainee **May 2026 – Present**
MEPCO, Multan

- Supporting Advanced Metering Infrastructure (AMI) operations, smart metering systems, and Meter Data Management System (MDMS) activities at MEPCO Head Office.
- Assisting in meter data monitoring, validation, analysis, and reporting to enhance operational efficiency and system performance.
- Supporting smart meter communication, system integration, troubleshooting, and coordination with field teams during deployment and testing.
- Gaining practical exposure to metering, protection, and instrumentation systems through technical analysis and operational support activities.

Management Trainee Engineer **Jun 2025 – May 2026**
Volka Food International, Multan

- Designed, tested, and troubleshoot electronic circuits and automation systems under senior engineers' supervision
- Performed preventive maintenance, root cause analysis, and process automation, reducing unplanned downtime and improving equipment reliability
- Hands-on PLC programming (Siemens & Allen-Bradley), HMI development, and electrical safety implementation
- Prepared technical documentation, specifications, and compliance reports

Intern **July 2024 – Aug 2024**
MEPCO, Multan

- Participated in installation, calibration, and maintenance of power distribution equipment and control systems
- Received practical training in the Metering & Testing (M&T) Department, including energy meter inspection, meter installation, testing, calibration procedures, and meter accuracy verification.
- Gained hands-on exposure in the Protection & Instrumentation (P&I) Department, learning the operation and maintenance of protection relays, control panels, and substation instrumentation systems.
- Assisted in technical fault analysis and preparation of progress reports

PUBLICATIONS

- **Advanced EEG-based brain monitoring for early detection of harmful neural activities using EfficientNet architecture**, *Biomedical Signal Processing and Control*, 2025. DOI: https://doi.org/10.1016/j.bspc.2025.108360

PROJECT

Brain Monitoring System using EEG Signals (Final Year Project) – NFC-IET, Multan

- Designed a deep learning model for EEG-based detection of harmful brain activity.
- Utilized Python and EfficientNet-B0 for CNN-based EEG signal analysis to assist neurological diagnosis.

Semester Projects – NFC-IET, Multan

- Solar Tracker System | Electronic Dice (Arduino) | Touch Sensor Implementation
- Clap-Activated Switch | 12V to 5V DC Converter

SKILLS & CORE COMPETENCIES

- **Technical & Engineering Skills** | Circuit Analysis | PCB Design | PLC Programming | Proteus Simulation | MATLAB | Electronics Troubleshooting | Technical Documentation | System Calibration | Control Systems
- **Project & Administrative Skills** | Project Planning | Technical Report Preparation | Workflow Optimization | Team Coordination | Research & Analysis | Process Improvement | Resource Management | Equipment Maintenance